THE LARGEST ADMISSIBLE DIGIT SETS MODULO PRIMES UP TO 31

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ABSTRACT. Let $r_k(\mathbb{Z}_m^n)$ denote the largest size of a subset of $\mathbb{Z}_m^n$ containing no arithmetic progression of length k. Elsholtz, Klahn and Lipnik construct large progression-free sets from *admissible digit sets* $D \subseteq \mathbb{Z}_p$ and tabulate, for primes $p \le 31$ and $3 \le k \le 8$, the largest admissible D they could find. Seventeen entries of their table are recorded only as lower bounds. We determine all of them. Two entries increase: the largest admissible digit sets modulo 31 have size 23 for $k = 5$ and 24 for $k = 6$, rather than at least 22 and 23, so that

$$r_5(\mathbb{Z}_{31}^n) \gg \frac{23^n}{n^{11}}, \qquad r_6(\mathbb{Z}_{31}^n) \gg \frac{24^n}{n^{23/2}}.$$

The other fifteen tabulated values are optimal. For three-term progressions we prove that the largest admissible digit set modulo any odd prime p has exactly $(p+1)/2$ elements, so the digit-set construction cannot improve the exponential base $(p+1)/2$ for $k = 3$.

The tool is an observation about the decision problem. Admissibility is the non-existence of a non-zero non-negative integral vector in a rational subspace, and has accordingly been treated as an integer program. The feasible set is a rational cone, so it contains a non-zero integral point precisely when it contains a non-zero real one, and admissibility is a linear program. Gordan's theorem then supplies a short certificate of admissibility, verifiable in exact integer arithmetic, and non-admissibility is certified by an explicit integral solution. Every lower bound in the table is certified in this way, and every upper bound is either a consequence of the three-term theorem or certified by a list of non-admissible sets with a symmetry-free covering check.

1. INTRODUCTION

For integers m, n, k let $r_k(\mathbb{Z}_m^n)$ denote the maximal size of a subset of $\mathbb{Z}_m^n := (\mathbb{Z}/m\mathbb{Z})^n$ containing no arithmetic progression of length k consisting of k distinct elements. For $k = 3$ the problem has a long history, going back to Behrend [1], Roth [9] and Szemerédi [11], and the polynomial method of Croot, Lev and Pach [2] and Ellenberg and Gijswijt [3] gave the decisive upper bounds in the finite-field setting. Lower bounds for general m and general progression length k are much less developed. Lin and Wolf [8] proved that for a prime p and $k \le p$,

$$r_k(\mathbb{Z}_p^n) \;\ge\; \left(p^{2(k-1)} + p^{k-1} - 1\right)^{n/2k},$$

and Elsholtz and Pach [7], followed by Elsholtz, Klahn and Lipnik [4], improved the base of the exponential by a Behrend-type digit construction, which we now recall.

Definition 1.1. Let p be a prime, let $D \subseteq \mathbb{Z}_p$ with $\ell := |D|$, and let n be a positive integer divisible by ℓ. Put

$$S(D,n) := \Big\{(v_1,\dots,v_n) \in \mathbb{Z}_p^n : \text{ every } d \in D \text{ occurs exactly } n/\ell \text{ times among } v_1,\dots,v_n\Big\}.$$

We call D *admissible for* k if $S(D,n)$ contains no arithmetic progression of length k consisting of k distinct elements, for every such n.

Since $|S(D,n)|$ is a central multinomial coefficient, an admissible D yields a lower bound whose exponential base is $|D|$ (Lemma 2.1 below), so the quantity of interest is

$$\delta_k(p) := \max\{\,|D| : \; D \subseteq \mathbb{Z}_p \text{ admissible for } k\,\}.$$

Table 1 of [4] records $\delta_k(p)$ for primes $5 \le p \le 31$ and $3 \le k \le 8$. For $p \in \{23, 29, 31\}$ seventeen entries are given only as lower bounds, the authors noting that they list "a lower bound for it if the existence of larger digit sets cannot be excluded". In particular [4] records $\delta_5(31) \ge 22$ and $\delta_6(31) \ge 23$, realised by the digit sets $[0, 17] \cup \{19, 20, 26, 29\}$ and $[0, 18] \cup \{20, 26, 29, 30\}$. Here and below $[a, b]$ denotes $\{x \in \mathbb{Z} : a \le x \le b\}$, regarded as a subset of $\mathbb{Z}_p$.

Throughout, p is a prime and k is an integer with $3 \le k < p$. (For $k \ge p$ the problem is degenerate, see Remark 2.5.)

Results. The first result is the observation that drives everything else. Write $\mathrm{P}_k(D)$ for the set of non-constant k-term arithmetic progressions $v = (v_1, \dots, v_k)$ in $\mathbb{Z}_p$ with all $v_i \in D$, regarded as ordered k-tuples, so that a progression and its reversal are distinct elements of $\mathrm{P}_k(D)$.

Theorem 1.2. *D is admissible for k if and only if there exist real numbers $w(d, i)$, for $d \in D$ and $1 \le i \le k$, such that*

$$\sum_{i=1}^{k} w(d, i) = 0 \quad (d \in D), \qquad \sum_{i=1}^{k} w(v_i, i) > 0 \quad (v \in \mathrm{P}_k(D)). \tag{1}$$

Moreover w may be taken rational, and hence integral.

A certificate satisfying (1) is a finite list of integers whose verification requires only summing $|D| + |\mathrm{P}_k(D)|$ integer expressions. Conversely, non-admissibility is certified by a non-negative integral vector indexed by $\mathrm{P}_k(D)$ (Lemma 2.2). Both kinds of certificate are checked by the script of Appendix B in exact arithmetic. This makes exhaustive searches practical and their outcomes verifiable, and it also yields a clean structural statement for three-term progressions.

Theorem 1.3. *For every odd prime p, $\delta_3(p) = (p + 1)/2$.*

Theorem 1.4. *Let $p \le 31$ be a prime and $3 \le k < p$. Then $\delta_k(p)$ is the entry of Table 1. In particular the seventeen entries recorded in* [4, Table 1] *only as lower bounds are determined: fifteen of them are attained by the digit sets listed in* [4]*, while*

$$\delta_5(31) = 23 \qquad \text{and} \qquad \delta_6(31) = 24,$$

attained by

$$D_5 := [0, 23] \setminus \{3\} \qquad \text{and} \qquad D_6 := [0, 25] \setminus \{1, 12\}.$$

Combining Theorem 1.4 with Lemma 2.1 gives the following.

Corollary 1.5. *As $n \to \infty$,*

$$r_5(\mathbb{Z}_{31}^n) \gg \frac{23^n}{n^{11}}, \qquad r_6(\mathbb{Z}_{31}^n) \gg \frac{24^n}{n^{23/2}},$$

improving the bases 22 and 23 obtainable from the digit sets of [4]*. For every prime $p \le 31$ and $3 \le k < p$, the entry of Table 1 is the largest exponential base that Definition 1.1 can produce for $r_k(\mathbb{Z}_p^n)$, and for $k = 3$ the base $(p + 1)/2$ is the largest one for every odd prime p.*

Theorem 1.3 explains a barrier. The bound $r_3(\mathbb{Z}_p^n) \gg ((p + 1)/2)^n / n^{(p-1)/4}$ follows from the interval $[0, (p - 1)/2]$, which is the $k = 3$ column of [4, Table 1], and no other choice of D does better. Elsholtz, Proske and Sauermann [5] have recently obtained bases exceeding $p/2$ for $k = 3$ by a genuinely different, two-dimensional construction, and Theorem 1.3 shows that such a change of construction was necessary.

The paper is organised as follows. Section 2 sets up the construction and proves the lemmas underlying it, including the trivial upper bound $\delta_k(p) \le p - 2$. Section 3 proves Theorem 1.2 and records a symmetric form of the certificate. Section 4 proves Theorem 1.3. Section 5 treats the

$p\backslash k$	3	4	5	6	7	8
5	3	3	·	·	·	·
7	4	5	5	5	·	·
11	6	7	8	9	9	9
13	7	8	10	11	11	11
17	9	10	13	13	15	15
19	10	11	14	15	16	17
23	12	13	17	18	19	20
29	15	17	21	22	24	25
31	16	18	**23**	**24**	26	26

TABLE 1. The values $\delta_k(p)$ for $3 \le k < p \le 31$. Entries for $p \le 19$ and $(p,k) = (23,6)$ were already stated as exact in [4, Table 1]. The two entries in bold improve the lower bounds 22 and 23 of [4]. The remaining fifteen entries with $p \ge 23$ coincide with the lower bounds of [4], which are thus optimal. Dots mark the degenerate range $k \ge p$.

two digit sets modulo 31, Section 6 describes the exhaustive search and the certificates behind the upper bounds of Theorem 1.4, and Section 7 collects remarks and open questions. Appendix A prints the certificates for D_5 and D_6, Appendix B the verification script, and Appendix C the covering-check program. All certificates and the search code accompany the paper as ancillary files.

2. The digit construction

Throughout, $D \subseteq \mathbb{Z}_p$ with $\ell := |D| \ge 2$.

Lemma 2.1. *Let $\ell \mid n$. Then*

$$|S(D,n)| = \binom{n}{n/\ell,\dots,n/\ell} \sim c_\ell \frac{\ell^n}{n^{(\ell-1)/2}} \qquad (n \to \infty),$$

where $c_\ell = \ell^{\ell/2}(2\pi)^{-(\ell-1)/2}$. Consequently, if D is admissible for k then $r_k(\mathbb{Z}_p^n) \gg_{p,k} \ell^n/n^{(\ell-1)/2}$ for all n.

Proof. A vector of $S(D,n)$ is determined by the partition of the n coordinate positions into the ℓ blocks on which it takes each of the digits of D, and each block has size n/ℓ. This gives the multinomial coefficient. By Stirling's formula $t! \sim \sqrt{2\pi t}\,(t/e)^t$, so with $t = n/\ell$,

$$\frac{n!}{((n/\ell)!)^\ell} \sim \frac{\sqrt{2\pi n}\,(n/e)^n}{(2\pi n/\ell)^{\ell/2}\,(n/(e\ell))^n} = \sqrt{2\pi n}\;\ell^{\,n}\;(2\pi n/\ell)^{-\ell/2} = \ell^{\ell/2}(2\pi)^{-(\ell-1)/2}\,\frac{\ell^n}{n^{(\ell-1)/2}},$$

since the factors $(n/e)^n$ and $(n/(e\ell))^n$ differ exactly by ℓ^n.

For the last assertion, if $\ell \nmid n$ write $n = \ell q + r$ with $0 \le r < \ell$ and embed $S(D, n-r)$ into $\mathbb{Z}_p^n$ by appending r zero coordinates. A k-term progression of distinct points in the image has all its r trailing coordinates constant, hence restricts to a k-term progression of k distinct points in $S(D,n-r)$. So the image is again progression-free, and its size is $|S(D,n-r)| \gg \ell^n/n^{(\ell-1)/2}$ because r is bounded. □

The next lemma is the combinatorial heart of the matter and is stated in [4, §4.1]. We include the short proof because both directions are used below, the second one to certify non-admissibility. For a non-constant k-term progression $v = (v_1,\dots,v_k)$ with entries in D we think

of v as a possible *coordinate pattern*: if $\mathbf{u}_1, \dots, \mathbf{u}_k$ is a progression in $\mathbb{Z}_p^n$ with common difference $\mathbf{c}$, then in each coordinate j the entries $\mathbf{u}_1^{(j)}, \dots, \mathbf{u}_k^{(j)}$ form a k-term progression in $\mathbb{Z}_p$ with common difference $\mathbf{c}^{(j)}$, and this pattern is constant precisely when $\mathbf{c}^{(j)} = 0$.

Lemma 2.2. *D fails to be admissible for k if and only if there exist non-negative integers $(x_v)_{v \in \mathrm{P}_k(D)}$, not all zero, satisfying*

$$\sum_{\substack{v \in \mathrm{P}_k(D) \\ v_i = d}} x_v \;=\; \sum_{\substack{v \in \mathrm{P}_k(D) \\ v_j = d}} x_v \qquad \text{for all } d \in D \text{ and all } 1 \le i < j \le k. \tag{2}$$

Proof. Suppose first that $S(D,n)$ contains a progression $\mathbf{u}_1, \dots, \mathbf{u}_k$ of k distinct points, with common difference $\mathbf{c} \neq 0$. For a non-constant pattern $v \in \mathrm{P}_k(D)$ let x_v be the number of coordinates $j \in \{1, \dots, n\}$ whose pattern $(\mathbf{u}_1^{(j)}, \dots, \mathbf{u}_k^{(j)})$ equals v. Every coordinate pattern lies in D^k because each $\mathbf{u}_i \in S(D,n)$, and is a k-term progression, so it is either constant or lies in $\mathrm{P}_k(D)$. Since $\mathbf{c} \neq 0$ some coordinate is non-constant, so not all x_v vanish. Now fix $d \in D$ and $1 \le i \le k$. The number of coordinates j with $\mathbf{u}_i^{(j)} = d$ equals

$$x_d^{\mathrm{const}} + \sum_{\substack{v \in \mathrm{P}_k(D) \\ v_i = d}} x_v,$$

where x_d^{const} counts the coordinates with constant pattern $(d, \dots, d)$. Note that x_d^{const} does not depend on i. As $\mathbf{u}_i \in S(D,n)$, this number equals n/ℓ for every i. Subtracting the expressions for two indices i and j gives (2).

Conversely, suppose non-negative integers x_v, not all zero, satisfy (2). For $d \in D$ write f_d for the common value in (2), and put $N := \sum_v x_v$. Summing f_d over $d \in D$ counts each v once, through the digit v_1, so $\sum_{d \in D} f_d = N$. Choose any integer $M \ge \max_d f_d$, set $x_d^{\mathrm{const}} := M - f_d \ge 0$, and put

$$n := \sum_{v \in \mathrm{P}_k(D)} x_v + \sum_{d \in D} x_d^{\mathrm{const}} = N + \ell M - N = \ell M,$$

so that $\ell \mid n$ and $n/\ell = M$. Partition $\{1, \dots, n\}$ into blocks of sizes x_v ($v \in \mathrm{P}_k(D)$) and x_d^{const} ($d \in D$), assign to the coordinates of the block of v the pattern v, and to those of the block of d the constant pattern $(d, \dots, d)$. Reading off the i-th entries defines vectors $\mathbf{u}_1, \dots, \mathbf{u}_k \in \mathbb{Z}_p^n$ which form an arithmetic progression, since they do so in each coordinate. By construction each digit $d \in D$ occurs in $\mathbf{u}_i$ exactly $x_d^{\mathrm{const}} + f_d = M = n/\ell$ times, for every i, so $\mathbf{u}_i \in S(D,n)$. Finally some $x_v > 0$ with v non-constant, so the common difference is non-zero, and as $k < p$ the k entries of a non-constant progression in $\mathbb{Z}_p$ are distinct, whence the $\mathbf{u}_i$ are k distinct points. Thus D is not admissible. □

Two consequences of Lemma 2.2 organise the search.

Lemma 2.3. (a) *If $D' \subseteq D$ and D is admissible for k, then so is D'.*
(b) *If D is admissible for k then so is $aD + b$ for every $a \in \mathbb{Z}_p^\times$ and $b \in \mathbb{Z}_p$.*

Proof. (a) A solution of (2) for D', extended by zeros to $\mathrm{P}_k(D) \supseteq \mathrm{P}_k(D')$, is a solution for D: the equations for $d \in D'$ are unchanged, and both sides of the equations for $d \in D \setminus D'$ vanish. (b) The map $x \mapsto ax + b$ is a bijection of $\mathbb{Z}_p$ carrying non-constant progressions to non-constant progressions, so it induces a bijection $\mathrm{P}_k(D) \to \mathrm{P}_k(aD + b)$ under which the systems (2) for D and for $aD + b$ correspond. □

Proposition 2.4. *$\delta_k(p) \le p - 2$ for all $3 \le k < p$, with equality if and only if $\mathbb{Z}_p \setminus \{0, 1\}$ is admissible for k.*

Proof. Let $D = \mathbb{Z}_p \setminus \{0\}$. For $d \in D$ and $1 \le i \le k$, the progressions $v \in \mathrm{P}_k(D)$ with $v_i = d$ are the tuples $(d + (j-i)c)_{1\le j\le k}$ with $c \neq 0$ and $d + (j-i)c \neq 0$ for all $j \neq i$, that is, with c avoiding the $k-1$ distinct non-zero values $d/(i-j)$, $j \neq i$. Hence there are exactly $p - k \ge 1$ of them, independently of d and i, and $x_v := 1$ for all $v \in \mathrm{P}_k(D)$ is a non-zero solution of (2). So D is not admissible, and by Lemma 2.3(a) neither is $\mathbb{Z}_p$, which gives $\delta_k(p) \le p-2$. Every $(p-2)$-subset of $\mathbb{Z}_p$ is an affine image of $\mathbb{Z}_p \setminus \{0,1\}$, since the affine group is transitive on pairs of distinct points, so by Lemma 2.3(b) equality holds exactly when this set is admissible. □

Table 1 shows equality in Proposition 2.4 for $(p,k) = (5,3)$, $(5,4)$, $(7,4)$, $(7,5)$, $(7,6)$, $(11,6)$, $(11,7)$, $(11,8)$, $(13,6)$, $(13,7)$, $(13,8)$, $(17,7)$, $(17,8)$ and $(19,8)$, and strict inequality otherwise.

Remark 2.5. For $k > p$ no progression in $\mathbb{Z}_p^n$ has k distinct elements, so $r_k(\mathbb{Z}_p^n) = p^n$ and $\mathbb{Z}_p$ itself is admissible. For $k = p$ a progression with non-zero difference has p distinct elements and covers all of $\mathbb{Z}_p$ in some coordinate, so $\mathbb{Z}_p \setminus \{0\}$ is admissible while $\mathbb{Z}_p$ is not (take $n = p$, $\mathbf{u}_1 = (0, 1, \dots, p-1)$ and $\mathbf{c} = (1, \dots, 1)$), whence $\delta_p(p) = p - 1$. This accounts for the entries $\delta_5(5) = 4$, $\delta_6(5) = \delta_7(5) = \delta_8(5) = 5$, $\delta_7(7) = 6$ and $\delta_8(7) = 7$ of [4, Table 1], which we omit from Table 1.

3. Admissibility is a linear program

Write A for the coefficient matrix of the homogeneous system (2), whose columns are indexed by $\mathrm{P}_k(D)$ and whose rows are indexed by triples (d,i,j) with $d \in D$ and $1 \le i < j \le k$. The entry of row (d,i,j) in column v is $[v_i = d] - [v_j = d]$. Lemma 2.2 says that D is admissible for k if and only if

$$C := \{\, x \in \mathbb{R}^{\mathrm{P}_k(D)} :\ Ax = 0,\ x \ge 0 \,\} \tag{3}$$

contains no non-zero *integral* point.

Proposition 3.1. *C contains a non-zero integral point if and only if $C \neq \{0\}$.*

Proof. One direction is trivial. For the other, suppose $C \neq \{0\}$ and consider the bounded polyhedron $Q := \{x :\ Ax = 0,\ 0 \le x \le 1\} \subseteq C$. Scaling a non-zero point of C shows that the linear function $x \mapsto \sum_v x_v$ is positive somewhere on Q, so its maximum over Q is positive and is attained at a vertex x^* of Q. A vertex of a polyhedron defined by rational data is the unique solution of a subsystem of the defining equations and inequalities, taken with equality, so x^* is rational (see [10, Ch. 8]). Multiplying x^* by a common denominator of its entries gives a non-zero integral point of C. □

Thus admissibility is decided by a linear program, and its infeasible instances have Farkas-type certificates. We use the homogeneous form due to Gordan: for a real matrix A, exactly one of the systems

$$\text{(i) } Ax = 0,\ x \ge 0,\ x \neq 0 \qquad \text{and} \qquad \text{(ii) } y^{\mathsf{T}}A > 0 \text{ (all entries strictly positive)}$$

is solvable [6], see also [10, Ch. 7].

Proof of Theorem 1.2. By Lemma 2.2 and Proposition 3.1, D is admissible for k if and only if system (i) above is unsolvable, that is, if and only if there is y with $y^{\mathsf{T}}A > 0$.

It remains to identify the linear functionals of the form $x \mapsto y^{\mathsf{T}}Ax$. The entry of $y^{\mathsf{T}}A$ in the column v is

$$\sum_{d\in D}\sum_{i<j} y_{d,i,j}\big([v_i = d] - [v_j = d]\big) = \sum_{i=1}^{k} w(v_i, i), \qquad w(d,i) := \sum_{j>i} y_{d,i,j} - \sum_{j<i} y_{d,j,i}.$$

These w satisfy $\sum_{i=1}^{k} w(d,i) = 0$ for each d, since every $y_{d,i,j}$ occurs once with each sign. Conversely, if $w(d,\cdot)$ is any vector in $\mathbb{R}^k$ with $\sum_i w(d,i) = 0$, then $w(d,\cdot)$ lies in the span of the vectors $e_i - e_j$, so it arises from some choice of $y_{d,i,j}$. Hence the achievable functionals are exactly $x \mapsto \sum_v x_v \sum_i w(v_i, i)$ with w ranging over the vectors satisfying the first half of (1), and the condition $y^{\mathsf{T}} A > 0$ is exactly the second half.

Finally, the system (1) is a finite system of strict and non-strict linear inequalities with rational coefficients. If it has a real solution it has a rational one, because the solution set is a non-empty relatively open subset of a rational subspace, in which the rational points are dense. Scaling by a common denominator gives an integral one, the conditions being homogeneous of degree one. □

Remark 3.2. The "reducibility" technique of [4, §4.2], in which one row-reduces A and repeatedly deletes columns meeting a row of constant sign, is a special way of producing a certificate w as in (1): each such row is a functional in the row space of A that is non-negative on all surviving columns and positive on the deleted ones, and a combination of these rows with rapidly decreasing positive weights is positive on every column. Theorem 1.2 shows that this family of arguments is complete: whenever D is admissible, some w works. In [4] the admissibility of four of the tabulated digit sets, those for $(p,k) = (17,7), (29,7), (31,7), (31,8)$, could not be established by reducibility and was checked by an integer-programming solver instead. Certificates (1) for admissible sets of these four sizes, among them the three sets of [4] for $p \in \{29, 31\}$, are included in the ancillary file `certificates.json`.

Certificates can be normalised. Reversal $v \mapsto \bar{v} := (v_k, \dots, v_1)$ is a bijection of $\mathrm{P}_k(D)$, so if w satisfies (1) then so does $\bar{w}(d,i) := w(d, k+1-i)$, and so does $(w + \bar{w})/2$.

Proposition 3.3. *If D is admissible for k, there is a certificate (1) with $w(d,i) = w(d,k+1-i)$ for all d and i. In particular:*

(a) *D is admissible for $k = 3$ if and only if there is $g\colon D \to \mathbb{R}$ with*

$$2g(m) > g(a) + g(b) \qquad \text{for every } (a,m,b) \in \mathrm{P}_3(D); \tag{4}$$

(b) *D is admissible for $k = 4$ if and only if there is $f\colon D \to \mathbb{R}$ with*

$$f(a) + f(b) > f(a') + f(b') \qquad \text{for every } (a,a',b',b) \in \mathrm{P}_4(D). \tag{5}$$

Proof. The first assertion was proved above. A symmetric certificate for $k = 3$ has $w(d,1) = w(d,3) = -w(d,2)/2$. With $g := w(\cdot,2)$ the second half of (1) reads $-g(a)/2 + g(m) - g(b)/2 > 0$, which is (4), and conversely any g satisfying (4) defines a certificate in this way. For $k = 4$ a symmetric certificate has $w(d,1) = w(d,4) =: f(d)$ and $w(d,2) = w(d,3) = -f(d)$, and (1) becomes (5). □

Condition (4) says that g is strictly concave along every three-term progression inside D. For an integer interval D the function $g(x) = -x^2$ works. Condition (5) is the corresponding fourth-order statement, satisfied by $f(x) = x^2$ on integer intervals since $a^2 + (a+3c)^2 - (a+c)^2 - (a+2c)^2 = 4c^2$.

4. Three-term progressions

Proof of Theorem 1.3. Let p be an odd prime and $h := (p-1)/2$.

Lower bound. Let $D = [0,h]$, so $|D| = (p+1)/2$. We claim that every $(a,m,b) \in \mathrm{P}_3(D)$ is an arithmetic progression of integers in $[0,h]$. Indeed, represent the common difference by an integer c with $0 < |c| \le h$. For an integer $a \in [0,h]$ the integer $a+c$ lies in $[-h, 2h] = [-h, p-1]$. If $a + c < 0$ its residue modulo p lies in $[h+1, p-1]$, outside D, so $a + c \in [0,h]$ as an integer, and the same argument applied to $m = a + c$ shows $b = m + c \in [0,h]$ as an integer. Now

$g(x) := -x^2$ satisfies (4), since $2g(m) - g(a) - g(b) = a^2 + b^2 - (a+b)^2/2 = (a-b)^2/2 > 0$. By Proposition 3.3(a), D is admissible for 3.

Upper bound. Let $D \subseteq \mathbb{Z}_p$ with $|D| \geq (p+3)/2$, and suppose $g\colon D \to \mathbb{R}$ satisfies (4). Choose $d_0 \in D$ with $g(d_0)$ minimal. The map $x \mapsto 2d_0 - x$ is a bijection of $\mathbb{Z}_p$, so $|D \cap (2d_0 - D)| \geq 2|D| - p \geq 3$. Hence there is $x \in D$, $x \neq d_0$, with $2d_0 - x \in D$. Since p is odd, x, d_0 and $2d_0 - x$ are distinct, so $(x, d_0, 2d_0 - x) \in \mathrm{P}_3(D)$ and (4) gives $2g(d_0) > g(x) + g(2d_0 - x) \geq 2g(d_0)$, a contradiction. By Proposition 3.3(a), D is not admissible for 3. □

The proof uses only that p is odd: with the same definitions, the largest digit set $D \subseteq \mathbb{Z}_m$ admissible for 3 has $(m+1)/2$ elements for every odd $m \geq 3$.

5. The two digit sets modulo 31

Proof of the lower bounds $\delta_5(31) \geq 23$ and $\delta_6(31) \geq 24$. Both statements are instances of Theorem 1.2. For $D_5 = [0, 23] \setminus \{3\} \subseteq \mathbb{Z}_{31}$ one computes $|\mathrm{P}_5(D_5)| = 164$, and the integer table w printed in Appendix A satisfies $\sum_{i=1}^{5} w(d, i) = 0$ for each of the 23 digits $d \in D_5$ and $\sum_{i=1}^{5} w(v_i, i) \geq 1$ for each of the 164 progressions $v \in \mathrm{P}_5(D_5)$. The entries of w are integers of absolute value at most 588. By Theorem 1.2, D_5 is admissible for $k = 5$.

For $D_6 = [0, 25] \setminus \{1, 12\} \subseteq \mathbb{Z}_{31}$ one computes $|\mathrm{P}_6(D_6)| = 146$, and the corresponding table in Appendix A, with entries of absolute value at most 522, satisfies $\sum_{i=1}^{6} w(d, i) = 0$ for each of the 24 digits and $\sum_{i=1}^{6} w(v_i, i) \geq 1$ for each of the 146 progressions. Hence D_6 is admissible for $k = 6$. □

Proof of Corollary 1.5. Apply Lemma 2.1 with $\ell = |D_5| = 23$ and with $\ell = |D_6| = 24$, giving exponents $(\ell - 1)/2 = 11$ and $23/2$ respectively. The optimality statements are Theorems 1.4 and 1.3. □

Remark 5.1. Both new sets are intervals with a small number of holes, whereas the sets recorded in [4, Tables 4 and 5] for these two entries, $[0, 17] \cup \{19, 20, 26, 29\}$ and $[0, 18] \cup \{20, 26, 29, 30\}$, consist of an interval together with a scattered tail near p. We verified that those two sets are indeed admissible, of sizes 22 and 23, so the comparison is between objects of the same kind. The longest admissible initial intervals $[0, L]$ modulo 31 have 22 elements for $k = 5$ and 23 for $k = 6$, so the holes in D_5 and D_6 are needed. Since $\delta_5(31) = 23$ and $\delta_6(31) = 24$ by Theorem 1.4, both sets are maximal: no digit of $\mathbb{Z}_{31}$ can be added to either without destroying admissibility.

6. Exhaustive search and the upper bounds

6.1. **What is certified.** Fix p and k, and let N be the entry of Table 1. The lower bound $\delta_k(p) \geq N$ is certified, for every one of the 48 pairs (p, k) in the table, by an explicit admissible N-set with an integral certificate (1). The sets are those of [4, Tables 2–7] for $p \geq 23$ together with D_5 and D_6, and sets found by our own search for $p \leq 19$. For $k = 3$ the upper bound is Theorem 1.3. For every other pair the upper bound $\delta_k(p) \leq N$ is certified by a finite list $\mathcal{O}_{p,k}$ of subsets of $\mathbb{Z}_p$ together with, for each $O \in \mathcal{O}_{p,k}$, non-negative integers $(x_v)_{v \in \mathrm{P}_k(O)}$, not all zero, satisfying (2), and by the following covering statement:

(6) every $(N+1)$-subset of $\mathbb{Z}_p$ contains $aO + b$ for some $O \in \mathcal{O}_{p,k}$, $a \in \mathbb{Z}_p^\times$, $b \in \mathbb{Z}_p$.

Given these data the proof of $\delta_k(p) \leq N$ is immediate: each $O \in \mathcal{O}_{p,k}$ is non-admissible by Lemma 2.2, hence so is every affine image of O by Lemma 2.3(b) and every superset of such an image by Lemma 2.3(a), which by (6) includes every $(N+1)$-subset of $\mathbb{Z}_p$. The lists $\mathcal{O}_{p,k}$ and the vectors x are in the ancillary file `obstructions.json`, and the script of Appendix B checks the equations (2) for every listed set in exact integer arithmetic. Statement (6) is checked by the program of Appendix C, which uses no symmetry reduction at all: it marks the $p(p-1)$

affine images of every $O \in \mathcal{O}_{p,k}$ in a bitmap indexed by the 2^p subsets of $\mathbb{Z}_p$, closes the bitmap upwards under supersets by a single pass in increasing order of bitmask, and then inspects each of the $\binom{p}{N+1}$ subsets of size $N+1$. For $p = 31$ this takes about half a minute and 2^{31} bits of memory.

Proof of Theorem 1.4. Run the script of Appendix B on the ancillary files, and the program of Appendix C on each list $\mathcal{O}_{p,k}$. All checks pass. By Theorem 1.2 the lower bounds hold, by the argument above and Theorem 1.3 the upper bounds hold. □

Table 2 summarises the certificates for $p \in \{23, 29, 31\}$. The corresponding data for $p \le 19$ is in the ancillary files and re-establishes the 30 exact entries of [4] in the range $3 \le k < p \le 19$ independently.

p	k	$\delta_k(p)$	[4]	$\lvert\mathcal{O}_{p,k}\rvert$	sizes of O	affine images	$(N+1)$-subsets
23	3	12	≥ 12	5	7–8	2,277	1,144,066
23	4	13	≥ 13	10	8–12	4,554	817,190
23	5	17	≥ 17	25	16–18	11,638	33,649
23	6	18	18	10	18	4,554	8,855
23	7	19	≥ 19	3	19–20	1,012	1,771
23	8	20	≥ 20	1	21	253	253
29	3	15	≥ 15			Theorem 1.3	
29	4	17	≥ 17	6	8–16	4,060	34,597,290
29	5	21	≥ 21	267	18–21	215,586	1,560,780
29	6	22	≥ 22	193	20–23	153,874	475,020
29	7	24	≥ 24	12	24–25	8,932	23,751
29	8	25	≥ 25	2	25	1,218	3,654
31	3	16	≥ 16			Theorem 1.3	
31	4	18	≥ 18	36	9–15	33,480	141,120,525
31	5	23	≥ 22	180	18–22	166,160	2,629,575
31	6	24	≥ 23	139	18–24	127,100	736,281
31	7	26	≥ 26	17	25–26	15,345	31,465
31	8	26	≥ 26	22	26–27	17,980	31,465

TABLE 2. Certificates for the upper bounds for $p \in \{23, 29, 31\}$. For each pair, $\lvert\mathcal{O}_{p,k}\rvert$ non-admissible sets O (with the range of their sizes) suffice. The last two columns give the number of distinct affine images of these sets and the number of $(N+1)$-subsets of $\mathbb{Z}_p$, each of which contains one of the images. For $k = 3$ the upper bound is Theorem 1.3. For $p = 23$ the listed sets give an independent computational proof.

6.2. **How the lists were found.** The correctness of Theorem 1.4 rests only on the checks just described. The lists $\mathcal{O}_{p,k}$ were produced as follows. By Lemma 2.3, an $(N+1)$-subset D of $\mathbb{Z}_p$ need only be tested up to affine equivalence, and it is convenient to enumerate the complements: the group $\{x \mapsto ax + b\}$ of order $p(p-1)$ acts on the $(p-N-1)$-subsets of $\mathbb{Z}_p$, and we enumerate the orbit representatives whose bitmask is minimal in their orbit, by running through all subsets and rejecting a subset as soon as some affine image has a smaller bitmask. For $(p, k) = (31, 4)$ this leaves 151,924 representatives among the $\binom{31}{12} = 141{,}120{,}525$ complements. Each representative D is tested by the linear program of Section 3. If the linear program finds a non-zero point of the cone (3), D is not admissible. We then shrink D greedily, deleting one digit at a time as long as the set remains non-admissible, and record the resulting set

O in $\mathcal{O}_{p,k}$ together with all $p(p-1)$ affine images of O as bitmasks. Subsequent representatives that contain one of the recorded bitmasks are non-admissible by Lemma 2.3 and are skipped without solving a linear program. In every case the enumeration terminated without finding an admissible $(N+1)$-set. For $k=4$ the sets O are small (between 8 and 16 digits) and the filter is very effective: only 36 linear programs were solved for $(31,4)$ and 6 for $(29,4)$. For $k \geq 5$ the minimal non-admissible sets are large, often only one to three digits smaller than $N+1$ and never more than seven, and between 139 and 267 linear programs were needed for each of $(29,5)$, $(29,6)$, $(31,5)$ and $(31,6)$. The integral vectors x certifying each O were then obtained by a bounded mixed-integer program (minimising $\sum_v x_v$ subject to (2), $\sum_v x_v \geq 1$ and $0 \leq x_v \leq 16$) or, when that program did not finish within a time limit, from a vertex of the polytope $\{x \geq 0 : \ Ax = 0, \ \sum_v x_v = 1\}$, recomputed in exact rational arithmetic on its support and scaled to integers. The second method produces valid but large coefficients, up to $4 \cdot 10^{18}$ for some sets with $k \in \{7,8\}$, which is why the verification script uses arbitrary-precision integers. The certificates (1) for the lower-bound sets were found by a linear program minimising $\max |w(d,i)|$ subject to (1) with margin 1, followed by scaling and rounding. All floating-point output was discarded after the exact check. Only the integers survive.

6.3. **Validation.** The same procedure, run from scratch on all pairs with $3 \leq k < p \leq 19$, reproduces every exact entry of [4, Table 1] in that range. The criterion of Lemma 2.2 was also validated directly for $p \in \{5,7\}$ and every $D \subseteq \mathbb{Z}_p$ with $|D| \geq 2$: for each non-admissible D the progression in $S(D,n)$ built from a witness in the proof of Lemma 2.2 was constructed explicitly and checked, and for each admissible D the sets $S(D,n)$ with $n \leq 6$ were enumerated and found to be progression-free.

7. Remarks and open questions

Remark 7.1. For $p \geq 37$ the table of [4] has no entries, and nothing is known beyond the general theorems of [4] and Proposition 2.4. The methods of this paper apply verbatim. The cost is dominated by the enumeration of the affine orbits of $(N+1)$-subsets, of which there are about $\binom{p}{N+1}/p(p-1)$, and by the linear programs that the obstruction filter does not eliminate, whose number grows with the size of the minimal non-admissible sets. We have not attempted $p \geq 37$.

Remark 7.2. The largest admissible initial interval $[0,L]$ is often, but not always, optimal. For $p \in \{23,29,31\}$ its size equals $\delta_k(p)$ for $k \in \{3,7,8\}$ and for $(p,k) = (23,4), (23,5), (29,5)$, and falls short by one for $(23,6)$, $(29,6)$, $(31,4)$, $(31,5)$, $(31,6)$ and by two for $(29,4)$. Certificates for these interval computations are in the ancillary file `intervals.json`. The general theorems of [4] give $\delta_k(p) \geq \lfloor \frac{k-1}{k+1} p \rfloor + 1$ for odd $k \geq 5$ and $\delta_k(p) \geq \lfloor \frac{k-2}{k} p \rfloor + 2$ for even $k \geq 4$ with $p \equiv -1 \pmod{k}$, for every prime $p > k$, and the improvements reported here concern specific small parameters only and do not affect those theorems. Note that $31 \equiv 1 \pmod{6}$, so the even case does not apply to $\delta_6(31)$ at all.

Remark 7.3. Since a k-term progression contains a $(k-1)$-term progression among its first $k-1$ entries, a set admissible for $k-1$ is admissible for k, so $\delta_k(p)$ is non-decreasing in k. Table 1 is consistent with this, and shows that the increase from k to $k+1$ can be zero even for $k < p-2$, for instance $\delta_7(31) = \delta_8(31) = 26$ and $\delta_5(17) = \delta_6(17) = 13$.

Remark 7.4. Proposition 3.3 suggests a question. For $k=3$ the certificate reduces to a single function on D and Theorem 1.3 follows from an extremal argument. For $k=4$ the certificate reduces to a single function f satisfying (5), but no comparable closed form for $\delta_4(p)$ is visible in the data: $\delta_4(p) - (p+1)/2$ equals $0,1,1,1,1,1,1,2,2$ for $p = 5,7,\ldots,31$. A structural description of the minimal non-admissible sets for $k=4$, which have between 8 and 16 digits in the cases computed, might yield an exact formula.

Remark 7.5. Subsequent work of Elsholtz, Proske and Sauermann [5] gives lower bounds for three-term-progression-free sets in $\mathbb{F}_p^n$ with exponential base larger than $p/2$, by a two-dimensional lifting construction rather than through digit sets in $\mathbb{Z}_p$. Theorem 1.3 shows that no digit set in the sense of Definition 1.1 can achieve this. That work does not address $\delta_k(p)$ for $k \geq 4$.

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Appendix A. The certificates for D_5 and D_6

Each table lists integers $w(d, i)$ for d in the digit set and $1 \leq i \leq k$. In both cases every row sums to zero, and every $v \in \mathrm{P}_k(D)$ satisfies $\sum_i w(v_i, i) \geq 1$.

$p = 31$, $k = 5$, $D_5 = [0, 23] \setminus \{3\}$, $|\mathrm{P}_5(D_5)| = 164$.

d	$w(d,1)$	$w(d,2)$	$w(d,3)$	$w(d,4)$	$w(d,5)$
0	−347	188	445	188	−474
1	−224	220	138	220	−354
2	37	320	−588	320	−89
4	134	−185	229	−185	7
5	−120	−71	511	−71	−249
6	−195	53	411	53	−322
7	4	−233	588	−233	−126
8	−112	−7	365	−7	−239
9	−382	228	436	228	−510
10	−35	63	72	63	−163
11	−364	415	25	415	−491
12	−460	588	−128	588	−588
13	−109	338	−330	338	−237
14	−329	474	−162	474	−457
15	−237	459	−315	459	−366
16	−43	112	−9	112	−172
17	−31	188	−186	188	−159
18	−179	159	168	159	−307
19	284	−181	−77	−181	155
20	21	−86	259	−86	−108
21	166	−34	−136	−34	38
22	265	−127	−148	−127	137
23	293	−25	−407	−25	164

$p = 31$, $k = 6$, $D_6 = [0, 25] \setminus \{1, 12\}$, $|\mathrm{P}_6(D_6)| = 146$.

d	$w(d,1)$	$w(d,2)$	$w(d,3)$	$w(d,4)$	$w(d,5)$	$w(d,6)$
0	−498	425	173	160	80	−340
2	74	250	157	−38	−67	−376
3	268	−257	401	−230	−311	129
4	−162	171	206	−234	192	−173
5	−130	232	404	−362	322	−466
6	−175	522	−23	−522	522	−324
7	−3	432	−252	−201	205	−181
8	−367	513	67	59	250	−522
9	283	−73	−55	23	−511	333
10	155	−104	−15	−102	−250	316
11	222	30	−89	−102	33	−94
13	−223	200	277	20	26	−300
14	−174	296	306	38	−333	−133
15	130	304	−239	−111	83	−167
16	−63	−283	518	176	−163	−185
17	217	−224	321	−14	−365	65
18	452	−522	188	46	−522	358
19	80	−279	522	35	−249	−109
20	437	−155	128	−431	−16	37
21	139	123	−144	−512	311	83
22	175	181	−40	−403	181	−94
23	288	420	−522	−522	104	232
24	357	64	−158	−380	−405	522
25	517	−96	−126	−472	−233	410

Appendix B. Verification script

The following program uses only the Python standard library and exact integer arithmetic. It reads the certificates from the ancillary files `certificates.json` (admissible sets with tables w), `obstructions.json` (non-admissible sets with vectors x) and `intervals.json` (the longest admissible initial intervals, with a certificate for $[0, L]$ and a witness for $[0, L+1]$) and checks the conditions (1) and (2). Run with `--export` it writes the lists $\mathcal{O}_{p,k}$ in the plain-text format read by the program of Appendix C.

```
"""Exact verification of the certificates.  Standard library only.

Lower bounds (certificates.json): for each record, D is admissible for k by Theorem 1.2,
because the printed integers w(d,i) satisfy
    sum_i w(d,i) = 0      for every d in D,
    sum_i w(v_i,i) > 0    for every v in P_k(D).
Upper bounds (obstructions.json): for each record, every listed set O is not admissible
for k by Lemma 2.2, because the listed non-negative integers x_v (v in P_k(O)), not all
zero, give each digit of O the same count in every position.  The covering check that
every (N+1)-subset of Z_p contains an affine image of a listed set is done by cover.c.
Intervals (intervals.json): [0,L] admissible (certificate w) and [0,L+1] not (witness x).
"""
import json, sys


def progressions(D, p, k):
    """Non-constant k-term APs (ordered k-tuples) with all entries in D."""
    Ds, out = set(D), []
    for a in sorted(Ds):
        for c in range(1, p):
            v = tuple((a + i * c) % p for i in range(k))
            if all(t in Ds for t in v):
                out.append(v)
    return out


def check_certificate(rec):
    p, k, D = rec["p"], rec["k"], rec["D"]
    assert len(D) == len(set(D)) and all(0 <= d < p for d in D)
    w = {}
    for key, val in rec["w"].items():
        d, i = key.split(",")
        w[(int(d), int(i))] = int(val)
    for d in D:
        assert sum(w[(d, i)] for i in range(1, k + 1)) == 0, ("row sum", p, k, d)
    P = progressions(D, p, k)
    for v in P:
        s = sum(w[(v[i - 1], i)] for i in range(1, k + 1))
        assert s > 0, ("progression", p, k, v, s)
    return len(D), len(P)


def check_witness(p, k, O, x):
    """x: list of [v, coefficient].  Exact check of system (2)."""
```

```
    P = set(progressions(O, p, k))
    xi = {}
    for v, c in x:
        v = tuple(v)
        assert v in P and c > 0, ("bad witness entry", p, k, O, v, c)
        xi[v] = xi.get(v, 0) + c
    assert xi, ("empty witness", p, k, O)
    for d in O:
        counts = [sum(c for v, c in xi.items() if v[i] == d) for i in range(k)]
        assert len(set(counts)) == 1, ("unbalanced digit", p, k, O, d, counts)


def export_for_cover(obs_path="obstructions.json"):
    """Write cover_p_k.txt (format: 'p N m', then one line 'size e1 ... es' per set) for cover.c."""
    for rec in json.load(open(obs_path)):
        name = "cover_%d_%d.txt" % (rec["p"], rec["k"])
        with open(name, "w") as f:
            f.write("%d %d %d\n" % (rec["p"], rec["N"], len(rec["obstructions"])))
            for item in rec["obstructions"]:
                f.write("%d %s\n" % (len(item["O"]), " ".join(map(str, item["O"]))))


def main(cert_path="certificates.json", obs_path="obstructions.json", int_path="intervals.json"):
    if cert_path == "--export":
        return export_for_cover()
    for rec in json.load(open(cert_path)):
        n, m = check_certificate(rec)
        print("admissible: p=%2d k=%d |D|=%2d |P_k(D)|=%3d  D=%s" % (rec["p"], rec["k"], n, m, rec["D"
    for rec in json.load(open(obs_path)):
        p, k, N = rec["p"], rec["k"], rec["N"]
        for item in rec["obstructions"]:
            check_witness(p, k, item["O"], item["x"])
        print("non-admissible sets verified: p=%2d k=%d N=%2d  %d sets (sizes %d..%d)" %
              (p, k, N, len(rec["obstructions"]), min(len(o["O"]) for o in rec["obstructions"]),
               max(len(o["O"]) for o in rec["obstructions"])))
    for rec in json.load(open(int_path)):
        p, k, L = rec["p"], rec["k"], rec["L"]
        assert rec["D"] == list(range(L + 1)) and rec["O"] == list(range(L + 2))
        check_certificate(rec)
        check_witness(p, k, rec["O"], rec["x"])
        print("interval: p=%2d k=%d  [0,%2d] admissible, [0,%2d] not" % (p, k, L, L + 1))
    print("All certificates and witnesses verified.")


if __name__ == "__main__":
    main(*sys.argv[1:])
```

Appendix C. Covering check

The following C program checks statement (6) for one pair (p, k) without any symmetry reduction, as described in Section 6. Compile with `gcc -O2 cover.c -o cover` and run as `./cover < cover_31_5.txt`.

```
/* cover.c: symmetry-free check of the covering claim behind an upper bound delta_k(p) <= N.
```

```
   Input (stdin): p, N, then the number m of obstruction sets, then each set as
   "size e1 e2 ... es".  For each set, all p(p-1) affine images are marked in a bitmap over
   the 2^p subsets of Z_p; the bitmap is then closed upwards under taking supersets, and
   finally every subset of size exactly N+1 is checked to be marked.
   Output: the number of (N+1)-subsets, the number marked, and OK / FAIL.
   Runs in O(2^p * p) time and 2^p bits of memory. */
#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>
#include <string.h>

int main(void) {
    int p, N, m;
    if (scanf("%d %d %d", &p, &N, &m) != 3) return 2;
    uint64_t total = 1ULL << p;
    uint8_t *bit = calloc(total / 8 + 1, 1);
    if (!bit) { fprintf(stderr, "alloc failed\n"); return 2; }
    #define GET(x) ((bit[(x) >> 3] >> ((x) & 7)) & 1)
    #define SET(x) (bit[(x) >> 3] |= (uint8_t)(1u << ((x) & 7)))
    long nimg = 0;
    for (int t = 0; t < m; t++) {
        int s; if (scanf("%d", &s) != 1) return 2;
        int el[64];
        for (int i = 0; i < s; i++) { if (scanf("%d", &el[i]) != 1) return 2; if (el[i] < 0 || el[i] >
        for (int a = 1; a < p; a++) for (int b = 0; b < p; b++) {
            uint64_t im = 0;
            for (int i = 0; i < s; i++) im |= 1ULL << ((a * el[i] + b) % p);
            if (!GET(im)) { SET(im); nimg++; }
        }
    }
    /* upward closure: process masks in increasing order; supersets have larger values */
    for (uint64_t x = 0; x < total; x++) {
        if (!GET(x)) continue;
        uint64_t rest = (total - 1) & ~x;
        while (rest) {
            uint64_t lowbit = rest & (~rest + 1);
            SET(x | lowbit);
            rest ^= lowbit;
        }
    }
    uint64_t count = 0, marked = 0;
    for (uint64_t x = 0; x < total; x++) {
        if (__builtin_popcountll(x) == N + 1) { count++; if (GET(x)) marked++; }
    }
    printf("p=%d N=%d obstruction sets=%d distinct images=%ld  (N+1)-subsets=%llu marked=%llu  %s\n",
           p, N, m, nimg, (unsigned long long)count, (unsigned long long)marked,
           count == marked ? "OK" : "FAIL");
    free(bit);
    return count == marked ? 0 : 1;
}
```